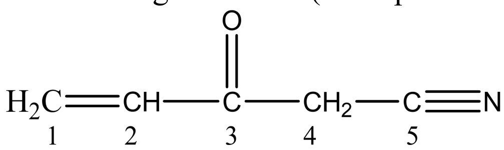
Chapter 9 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Consider the following molecule. (Lone pairs are not drawn in.)



Specify the hybridization of each carbon atom (in numeric order: C-1 C-2 C-3 C-4 C-5).
A) sp^2 sp^2 sp^3 sp^3 sp^3
B) sp^2 sp^2 sp^3 sp^3 sp
C) sp sp sp sp^2 sp
D) sp^2 sp^2 sp^2 sp^3 sp^3
E) sp^2 sp^2 sp^2 sp^3 sp

2. The molecule AX_3, in which A is the central atom, is polar and obeys the octet rule; therefore,
A) A has two lone pairs.
B) A has one lone pair.
C) A has no lone pairs.
D) A has four bonding pairs.
E) A has three lone pairs.

3. Which of the following has the shortest N—O bond?
A) NO_2^- B) N_2 C) NO^+ D) NO_3^- E) none of these

4. Which statement about the thiocyanate ion, NCS^-, is true?
A) Only one correct resonance structure can be drawn.
B) Its Lewis structure contains an unpaired electron.
C) Its shape is bent like that of H_2O.
D) There are more than two σ bonds in the ion.
E) none of these

5. According to MO theory, overlap of two *s* atomic orbitals produces ________.
A) one bonding molecular orbital and one hybrid orbital
B) two bonding molecular orbitals
C) two bonding molecular orbitals and two antibonding molecular orbitals
D) two bonding molecular orbitals and one antibonding molecular orbital
E) one bonding molecular orbital and one antibonding molecular orbital

6. Which of the following statements is <u>false</u>?
A) Paramagnetic molecules are attracted toward a magnetic field.
B) Paramagnetism cannot be deduced from the Lewis structure of a molecule alone.
C) Atoms or molecules with an even number of electrons are diamagnetic.
D) N_2 molecules are diamagnetic.
E) Atoms or molecules with an odd number of electrons are paramagnetic.

7. Three sulfur fluorides are observed: SF_2, SF_4, and SF_6.
Of these, ________ is/are polar.
A) SF_2
B) SF_2 and SF_4
C) SF_4
D) SF_6
E) SF_2, SF_4, and SF_6

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8. The carbon-hydrogen σ bond in ethylene, $CH_2{=\!=}CH_2$, results from the overlap of ________.
A) sp^2 hybrid orbital and *s* atomic orbital
B) *sp* hybrid orbitals
C) sp^3 hybrid orbitals
D) *s* hybrid orbital and sp^2 atomic orbital
E) *s* atomic orbitals

9. For how many of the following does the bond order decrease if you add one electron to the neutral molecule?
Be_2, C_2, N_2, O_2
A) 2　B) 1　C) 3　D) 0　E) 4

10. According to valence bond theory, which orbitals overlap in the formation of the bond in HCl?
A) 1*s* on H and 2*p* on Cl
B) 2*s* on H and 3*p* on Cl
C) 1*s* on H and 3*s* on Cl
D) 1*s* on H and 3*p* on Cl
E) 1*s* on H and 4*p* on Cl

11. Of the following, which molecule has the largest bond angle?
A) O_3　B) OF_2　C) HCN　D) H_2O
E) More than one of the above have equally large bond angles.

12. For molecules of the general formula AB_n, n can be greater than four ________.
A) for any element A
B) only when A is an element from period 3 and below in the periodic table
C) only when A is boron or beryllium
D) only when A is carbon
E) only when A is Xe

13. A typical triple bond ________.
A) consists of three shared electrons
B) consists of two σ bonds and one π bond
C) consists of six shared electron pairs
D) is longer than a single bond
E) consists of one σ bond and two π bonds

14. The more effectively two atomic orbitals overlap, ________.
A) the more bonding MOs will be produced by the combination
B) the higher will be the energy of the resulting bonding MO and the lower will be the energy of the resulting antibonding MO
C) the higher will be the energies of both bonding and antibonding MOs that result
D) the fewer antibonding MOs will be produced by the combination
E) the lower will be the energy of the resulting bonding MO and the higher will be the energy of the resulting antibonding MO

15. A molecule must have at least two resonance structures in order to display ________.
A) delocalized σ bonding
B) trigonal planar electron domain geometry
C) localized π bonding
D) delocalized π bonding
E) localized σ bonding

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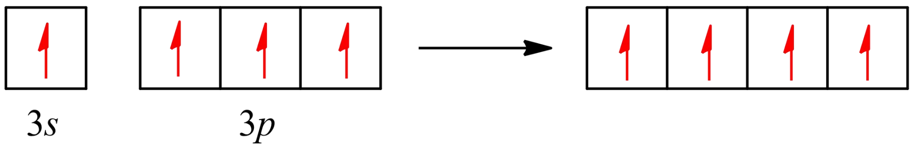
Part B Short questions (20 points)

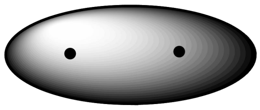
16. (7 points) **(a)** Explain why BrF_4^- is square planar, whereas BF_4^- is tetrahedral.
(b) How would you expect the H—X—H bond angle to vary in the series H_2O, H_2S, H_2Se? Explain. (*Hint:* The size of an electron pair domain depends in part on the electronegativity of the central atom.)

17. (3 points)The orbital diagram that follows presents the final step in the formation of hybrid orbitals by a silicon atom.



(a) Which of the following best describes what took place before the step pictured in the diagram: (i) Two 3*p* electrons became unpaired, (ii) An electron was promoted from the 2*p* orbital to the 3*s* orbital, or (iii) An electron was promoted from the 3*s* orbital to the 3*p* orbital?
(b) What type of hybrid orbital is produced in this hybridization?

18. (4 points) One of the molecular orbitals of the Li_2^- ion is sketched below:



(a) Is the molecular orbital a σ or π MO? Is it bonding or antibonding?

(b) In Li_2^-, how many electrons occupy the MO shown above?

(c) Compared to the Li—Li bond in Li_2, the Li—Li bond in Li_2^- is expected to be which of the following: (i) Shorter and stronger, (ii) longer and stronger, (iii) shorter and weaker, (iv) longer and weaker, or (v) the same length and strength?

19. (6 points) Explain the following: **(a)** B_2 is paramagnetic, which is consistent with the π_{2p} MOs being lower in energy than the σ_{2p} MO. **(b)** The O_2^{2+} ion has a stronger O—O bond than O_2 itself.

Chapter 9 Homework Answer Sheet

Name: **Student ID:** **Instructor:** **Score:**

Part A Multiple choice (15 points)

1-5______________

6-10_____________

11-15____________

Part B Short questions (20 points)